

Collision Ideals and Off-Diagonal Sheets

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Abstract

For a polynomial map $F = (F_1, \dots, F_n): \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n$, define in $S = \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n]$ the collision and diagonal ideals

$$I_R = (F_i(x) - F_i(y))_{i=1}^n, \quad I_{\Delta} = (x_i - y_i)_{i=1}^n.$$

The inclusion $I_R \subseteq I_{\Delta}$ defines the obstruction ideal $\text{Obs}(F) := I_{\Delta}/I_R \subseteq C_F := S/I_R$, and

$$\text{Obs}(F) = 0 \iff I_R = I_{\Delta} \iff F \text{ is a polynomial automorphism.}$$

In dimension two, let $\mathcal{D}_F$ be the finite-cover diagram of a map with constant nonzero Jacobian determinant, and let $\mathcal{H}(\mathcal{D}_F)$ be its hidden-inertia locus. Then

$$\mathcal{H}(\mathcal{D}_F) = \emptyset \implies I_R = I_{\Delta}.$$

For the separable nonnormal cubic class in dimension three, let

$$K = \mathbb{C}(F_1, F_2, F_3), \quad L = \mathbb{C}(x_1, x_2, x_3),$$

and let N be the normal closure of L/K . Then

$$L \otimes_K L \simeq L \times N, \quad \text{Gal}(N/K) \simeq S_3.$$

The off-diagonal factor N is the ordered-root Galois-closure sheet. Consequently, $I_{\Delta}/I_R \neq 0$. These constructions and implications are formalized in Lean 4, with the external inputs isolated explicitly.

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1 General construction

1.1 The two ideals and their obstruction

Throughout the paper, the base field is $\mathbb{C}$. Let

$$F = (F_1, \dots, F_n): X = \mathbb{A}_{\mathbb{C}}^n \longrightarrow \mathbb{A}_{\mathbb{C}}^n$$

be a polynomial map, and let

$$S = \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n].$$

The map F is Keller if

$$\det JF \in \mathbb{C}^\times.$$

Define

$$I_R(F) = (F_i(x) - F_i(y))_{i=1}^n, \quad I_\Delta = (x_i - y_i)_{i=1}^n, \quad \text{Obs}(F) := I_\Delta / I_R(F).$$

The quotient $\text{Obs}(F)$ is naturally an ideal of the collision ring $C_F := S / I_R(F)$; we call it the obstruction ideal. For a fixed map F , we abbreviate $I_R(F)$ to I_R .

Proposition 1.1 (Canonical containment). *For every polynomial map F ,*

$$I_R(F) \subseteq I_\Delta.$$

Proof. For $H \in \mathbb{C}[x_1, \dots, x_n]$, the difference $H(x) - H(y)$ belongs to $(x_1 - y_1, \dots, x_n - y_n)$. This follows by induction on H , or by telescoping each monomial one coordinate at a time. Apply the statement to each coordinate F_i . $\square$

Let $A := \mathbb{C}[x_1, \dots, x_n]$. Diagonal evaluation induces a surjective ring map

$$\bar{\mu}_F: C_F \longrightarrow A, \quad [s]_{I_R} \longmapsto s(x, x).$$

Proposition 1.2 (The diagonal kernel). *As ideals of C_F ,*

$$\ker(\bar{\mu}_F) = \text{Obs}(F) = I_\Delta / I_R.$$

Consequently,

$$\ker(\bar{\mu}_F) = 0 \iff \text{Obs}(F) = 0 \iff I_R = I_\Delta.$$

Proof. The kernel of diagonal evaluation $S \rightarrow A$ is I_Δ . Passing through S/I_R therefore identifies the displayed kernel with I_Δ/I_R . Since $I_R \subseteq I_\Delta$,

$$I_\Delta/I_R = 0 \iff I_R = I_\Delta.$$

□

1.2 The fiber-product interpretation

Let

$$A = \mathbb{C}[x_1, \dots, x_n], \quad B = \mathbb{C}[F_1, \dots, F_n] \subseteq A.$$

Let $X = \text{Spec } A$, let $Y = \text{Spec } B$, and let

$$f_F: X \longrightarrow Y$$

be the morphism induced by $B \hookrightarrow A$. The homomorphisms $\iota_x, \iota_y: A \rightrightarrows C_F$, defined by $\iota_x(a) = [a(x)]$ and $\iota_y(a) = [a(y)]$, agree on B because $[F_i(x)] = [F_i(y)]$ in C_F . Their common restriction equips C_F with a B -algebra structure.

Theorem 1.3 (Fiber-product interpretation). *There is a canonical B -algebra equivalence and its dual affine-scheme equivalence*

$$C_F \cong A \otimes_B A, \quad R_F := \text{Spec } C_F \cong X \times_Y X.$$

Under the first equivalence, $\bar{\mu}_F$ is tensor multiplication

$$A \otimes_B A \longrightarrow A, \quad a \otimes a' \longmapsto aa'.$$

Its dual morphism is the diagonal

$$\Delta_{X/Y}: X \longrightarrow X \times_Y X.$$

Proof. A map $C_F \rightarrow T$ to a commutative $\mathbb{C}$ -algebra T is equivalent to a pair of maps $f, g: A \rightrightarrows T$ satisfying

$$f(F_i) = g(F_i) \quad (1 \leq i \leq n).$$

Since the F_i generate B , this is precisely the universal property of $A \otimes_B A$. Applying Spec to this equivalence and using

$$\text{Spec}(A \otimes_B A) \cong \text{Spec } A \times_{\text{Spec } B} \text{Spec } A$$

gives the asserted fiber-product equivalence. The statements about multiplication and the diagonal follow on pure tensors. □

$$R_F(\mathbb{C}) = \{(u, v) \in X(\mathbb{C}) \times X(\mathbb{C}) : F(u) = F(v)\}.$$

Inside

$$X \times X = \text{Spec } S,$$

the ideals I_R and I_Δ define, respectively, $X \times_Y X$ and the image of $\Delta_{X/Y}$. The contravariance of $I \mapsto V(I)$ therefore gives the equivalences

$$I_R \subseteq I_\Delta \iff \Delta_{X/Y}(X) \subseteq X \times_Y X$$

and

$$I_R = I_\Delta \iff X \times_Y X = \Delta_{X/Y}(X)$$

as closed subschemes of $X \times X$. Thus the algebraic and geometric gaps are two descriptions of the same scheme-theoretic containment.

1.3 The off-diagonal factor and secant projectors

For every polynomial map F , define

$$I_{\text{off}} := I_R : I_\Delta, \quad C_F^\circ := S/I_{\text{off}}, \quad R_F^\circ := \text{Spec } C_F^\circ.$$

This definition does not require the diagonal to be clopen.

Corollary 1.4 (Off-diagonal emptiness criterion). *For every polynomial map F , the following are equivalent:*

$$R_F^\circ = \emptyset \iff C_F^\circ = 0 \iff \text{Obs}(F) = 0 \iff I_R = I_\Delta.$$

Proof.

$$R_F^\circ = \emptyset \iff C_F^\circ = 0 \iff I_R : I_\Delta = S.$$

Moreover,

$$I_R : I_\Delta = S \iff I_\Delta \subseteq I_R \iff I_R = I_\Delta.$$

The equivalence with $\text{Obs}(F) = 0$ is Proposition 1.2. □

The idempotent mechanism behind a clopen diagonal is purely commutative-algebraic. Let C be a commutative ring, $J \subseteq C$ an ideal, $\delta \in C$, and $c \in C^\times$. Suppose

$$\delta J = 0, \quad \delta - c \in J.$$

Define

$$q := 1 - c^{-1}\delta.$$

Theorem 1.5 (Abstract off-diagonal projector). *The element q is idempotent and*

$$J = Cq, \quad \text{Ann}_C(J) = C(1 - q).$$

In particular,

$$J = 0 \iff q = 0.$$

Proof. The congruence $\delta \equiv c \pmod{J}$ gives $q \in J$. For every $j \in J$,

$$qj = j - c^{-1}\delta j = j.$$

Taking $j = q$ proves $q^2 = q$; the same identity gives $J = Cq$. Finally,

$$aq = 0 \iff a = a(1 - q),$$

so $\text{Ann}_C(J) = \text{Ann}_C(Cq) = C(1 - q)$. □

Whenever the theorem is applied with

$$C = C_F, \quad J = \text{Obs}(F),$$

the identities $J = C_F q$ and $C_F/J \cong A$ give

$$C_F \cong A \times C_F/C_F(1 - q).$$

Thus the first factor is the diagonal and the second is its off-diagonal complement.

1.4 Generic fibers and marked-root decomposition

Let F be dominant and generically finite, and let

$$A = \mathbb{C}[x_1, \dots, x_n], \quad B = \mathbb{C}[F_1, \dots, F_n] \subseteq A.$$

Let

$$K = \text{Frac}(B), \quad L = \text{Frac}(A).$$

Let

$$\theta: K \otimes_B A \longrightarrow L, \quad \lambda \otimes a \longmapsto \lambda a.$$

Theorem 1.6 (Generic collision bridge). *The map θ is an isomorphism. Consequently, there is a K -algebra isomorphism*

$$K \otimes_B C_F \cong L \otimes_K L,$$

and the following square commutes:

$$\begin{array}{ccc} K \otimes_B C_F & \xrightarrow{1 \otimes \bar{\mu}_F} & K \otimes_B A \\ \sim \downarrow & & \downarrow \theta \sim \\ L \otimes_K L & \xrightarrow{\mu_L} & L. \end{array}$$

Proof. Let $U = B \setminus \{0\}$. Since $K = U^{-1}B$, localization gives

$$K \otimes_B A \xrightarrow{\sim} U^{-1}A, \quad \frac{b}{u} \otimes a \longmapsto \frac{ba}{u}.$$

Since $A = \mathbb{C}[x_1, \dots, x_n]$ is a domain and $B \subseteq A$,

$$B \setminus \{0\} \subseteq A \setminus \{0\}.$$

Hence

$$U^{-1}A \hookrightarrow \text{Frac}(A) = L, \quad \frac{a}{u} \longmapsto \frac{a}{u}.$$

Composing this injection with the displayed isomorphism gives θ , so θ is injective.

Since F is generically finite, L/K is a finite extension. The image of $U^{-1}A$ in L is

$$K[x_1, \dots, x_n].$$

Since each x_i is algebraic over K , this K -domain is finite-dimensional over K , hence is a field. Since it contains A , it contains $\text{Frac}(A) = L$; the reverse inclusion is already given by its embedding into L . Therefore

$$U^{-1}A = L,$$

and θ is an isomorphism.

By Theorem 1.3, identify C_F with $A \otimes_B A$. There is a K -algebra isomorphism

$$\beta: K \otimes_B (A \otimes_B A) \xrightarrow{\sim} (K \otimes_B A) \otimes_K (K \otimes_B A)$$

defined on pure tensors by

$$\beta(\lambda \otimes (a \otimes a')) = (\lambda \otimes a) \otimes (1 \otimes a').$$

Its inverse is

$$(\lambda \otimes a) \otimes (\lambda' \otimes a') \mapsto \lambda \lambda' \otimes (a \otimes a').$$

The balancing relations over B and K show that these formulas are well-defined, and the two displayed maps are mutually inverse. Composing β with $\theta \otimes_K \theta$ gives

$$\Phi_F: K \otimes_B C_F \xrightarrow{\sim} L \otimes_K L.$$

It remains to identify the diagonal map. Under $C_F \cong A \otimes_B A$, the homomorphism $\bar{\mu}_F$ is multiplication. Therefore, on a pure tensor,

$$\begin{aligned} (\mu_L \circ \Phi_F)(\lambda \otimes (a \otimes a')) &= (\lambda a) a' = \lambda a a', \\ (\theta \circ (1 \otimes \bar{\mu}_F))(\lambda \otimes (a \otimes a')) &= \theta(\lambda \otimes a a') = \lambda a a'. \end{aligned}$$

Pure tensors generate $K \otimes_B C_F$, so the two homomorphisms are equal. This proves the commutative square. $\square$

Suppose now that L/K is separable and has a power basis $L = K(\alpha)$. Let $m_\alpha(T) \in K[T]$ be the minimal polynomial of α , and let $h_\alpha(T) \in L[T]$ be determined by

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \quad \text{in } L[T].$$

Theorem 1.7 (Marked-root tensor decomposition). *Give $L \otimes_K L$ its L -algebra structure through the left tensor factor. There is an L -algebra equivalence*

$$L \otimes_K L \cong L \times L[T]/(h_\alpha),$$

under which tensor multiplication $L \otimes_K L \rightarrow L$ is projection onto the first factor.

Proof. The power-basis presentation

$$L \cong K[T]/(m_\alpha)$$

and base change from K to L give an L -algebra isomorphism

$$L \otimes_K L \xrightarrow{\sim} L[T]/(m_\alpha), \quad 1 \otimes \alpha \mapsto T.$$

Since m_α is separable, $m'_\alpha(\alpha) \neq 0$. Differentiating

$$m_\alpha(T) = (T - \alpha)h_\alpha(T)$$

and evaluating at $T = \alpha$ gives

$$h_\alpha(\alpha) = m'_\alpha(\alpha) \neq 0.$$

Thus $T - \alpha$ does not divide h_α , so these two polynomials are coprime in $L[T]$. The Chinese remainder theorem now gives

$$L[T]/(m_\alpha) \xrightarrow{\sim} L[T]/(T - \alpha) \times L[T]/(h_\alpha) \xrightarrow{\sim} L \times L[T]/(h_\alpha).$$

Under the first isomorphism above, a pure tensor $\ell \otimes p(\alpha)$, with $p \in K[T]$, corresponds to the class of $\ell p(T)$. Tensor multiplication sends it to $\ell p(\alpha)$, which is evaluation at $T = \alpha$. Under the Chinese remainder isomorphism, this evaluation map is projection onto the factor $L[T]/(T - \alpha) \cong L$. Hence tensor multiplication is the first projection. $\square$

Together with Theorem 1.6, this gives

$$K \otimes_B C_F \cong L \times L[T]/(h_\alpha),$$

where the first projection is the generic diagonal.

1.5 Normalized covers and inertia

Let L/K be finite and separable, and let N/K be its normal closure, with a fixed embedding $L \hookrightarrow N$. Let

$$G = \text{Gal}(N/K), \quad H = \text{Gal}(N/L).$$

Proposition 1.8 (Conjugate-embedding action). *Restriction induces a bijection*

$$G/H \xrightarrow{\sim} \text{Hom}_K(L, N), \quad gH \mapsto g|_L.$$

The induced action of G on G/H is faithful; equivalently,

$$\text{core}_G(H) = \bigcap_{g \in G} gHg^{-1} = 1.$$

Proof. For $g_1, g_2 \in G$,

$$g_1|_L = g_2|_L \iff g_2^{-1}g_1 \in H \iff g_1H = g_2H.$$

Every K -embedding $L \hookrightarrow N$ extends to an element of G , so restriction gives the stated bijection. Its action kernel is

$$\ker(G \curvearrowright G/H) = \bigcap_{g \in G} gHg^{-1} = \text{core}_G(H).$$

If $\sigma \in \text{core}_G(H)$, then

$$\sigma|_{g(L)} = \text{id}_{g(L)} \quad (g \in G).$$

Since N is the normal closure,

$$N = K(g(L) : g \in G),$$

and therefore $\sigma = \text{id}_N$. $\square$

Let

$$Y = \text{Spec } B, \quad \overline{X} = \text{Norm}_L(Y), \quad Z = \text{Norm}_N(Y).$$

The marked embedding $L \hookrightarrow N$ induces a canonical commutative triangle

$$Z \xrightarrow{\pi} \overline{X} \xrightarrow{\bar{\nu}} Y, \quad \nu = \bar{\nu} \circ \pi : Z \longrightarrow Y.$$

Suppose that $\bar{\nu}$ and ν are finite, and fix an open immersion over Y

$$j: X = \operatorname{Spec} A \hookrightarrow \bar{X}$$

with $\bar{\nu} \circ j = f_F$. Denote this diagram and its deleted boundary by

$$\mathcal{D} = \left(Z \xrightarrow{\pi} \bar{X} \xrightarrow{\bar{\nu}} Y, j: X \hookrightarrow \bar{X} \right), \quad \partial X = \bar{X} \setminus j(X).$$

Fix $E \in Z^{(1)}$. Let w_E be its divisorial valuation, let $b = \nu(E)$, and let $v_b = w_E|_K$. Let $D_E \leq G$ and $I_E \trianglelefteq D_E$ be its decomposition and inertia groups. Let

$$\mathcal{Q}_E := D_E \backslash G/H.$$

For $\omega = D_E g H \in \mathcal{Q}_E$, let

$$u_\omega = (g^{-1} w_E)|_L, \quad \operatorname{cent}(E, \omega) = \operatorname{cent}_{\bar{X}}(u_\omega),$$

and let

$$i_E(\omega) = [I_E : I_E \cap g H g^{-1}].$$

Proposition 1.9 (Conjugate divisors and double cosets). *The assignments above are well-defined, and restriction of valuations induces a bijection*

$$\mathcal{Q}_E \xrightarrow{\sim} \{\text{prime divisors of } \bar{X} \text{ above } b\}, \quad \omega \mapsto \operatorname{cent}(E, \omega).$$

Proof. The group G acts transitively on the extensions of v_b to N , and the stabilizer of w_E is D_E . Restricting such a valuation to $L = N^H$ identifies precisely those extensions whose indices differ on the right by an element of H . The resulting orbits are therefore indexed by $D_E \backslash G/H$, giving the stated bijection.

It remains to check independence of representatives. If $D_E g' H = D_E g H$, then $g' = d g h$ for some $d \in D_E$ and $h \in H$. The element d fixes w_E , while h fixes L pointwise, so

$$(g'^{-1} w_E)|_L = (g^{-1} w_E)|_L.$$

Moreover, $I_E \trianglelefteq D_E$, and conjugation by d identifies

$$I_E \cap g' H g'^{-1} \quad \text{with} \quad I_E \cap g H g^{-1}.$$

Thus u_ω , $\operatorname{cent}(E, \omega)$, and $i_E(\omega)$ are well-defined. □

We use the notation

$$e(\operatorname{cent}(E, \omega)/b) := e(u_\omega/v_b).$$

Since the residue characteristic is zero,

$$\operatorname{Ram}^{(1)}(Z/Y) := \{E \in Z^{(1)} : e(w_E/v_b) > 1\} = \{E \in Z^{(1)} : I_E \neq 1\}.$$

Proposition 1.10 (Detection of inertia on a conjugate sheet). *If $I_E \neq 1$, then there is an $\omega \in \mathcal{Q}_E$ with*

$$i_E(\omega) > 1.$$

Proof. Suppose, to the contrary, that $i_E(\omega) = 1$ for every $\omega \in \mathcal{Q}_E$. For each $g \in G$,

$$1 = i_E(D_E g H) = [I_E : I_E \cap g H g^{-1}],$$

so

$$I_E = I_E \cap g H g^{-1} \subseteq g H g^{-1} \quad \text{for every } g \in G.$$

Consequently,

$$I_E \subseteq \bigcap_{g \in G} g H g^{-1} = \text{core}_G(H) = 1,$$

contrary to $I_E \neq 1$. □

Proposition 1.11 (Intermediate ramification index). *For $\omega = D_E g H \in \mathcal{Q}_E$,*

$$e(\text{cent}(E, \omega)/b) = i_E(\omega) = [I_E : I_E \cap g H g^{-1}].$$

Proof. Let w_E and v_b be the divisorial valuations associated with E and b , and let

$$w_g = g^{-1} w_E, \quad u_\omega = w_g|_L.$$

The valuation u_ω has center $\text{cent}(E, \omega)$ on $\overline{X}$. Since the residue fields have characteristic zero,

$$e(w_g/v_b) = |I_E|.$$

The inertia group of w_g over u_ω is

$$g^{-1} I_E g \cap H,$$

and hence

$$e(w_g/u_\omega) = |g^{-1} I_E g \cap H| = |I_E \cap g H g^{-1}|.$$

Multiplicativity of ramification indices in the field tower

$$K \subseteq L \subseteq N$$

therefore gives

$$e(u_\omega/v_b) = \frac{e(w_g/v_b)}{e(w_g/u_\omega)} = [I_E : I_E \cap g H g^{-1}],$$

which is the asserted formula. □

Proposition 1.12 (Visibility-to-boundary). *Suppose that $X \rightarrow Y$ is étale. For $E \in Z^{(1)}$ and $\omega \in \mathcal{Q}_E$,*

$$i_E(\omega) > 1 \implies \text{cent}(E, \omega) \in \partial X.$$

Proof. Suppose that $\text{cent}(E, \omega) \in j(X)$, and let $x \in X$ be the corresponding point under the open immersion j . The local homomorphism

$$\mathcal{O}_{Y,b} \longrightarrow \mathcal{O}_{X,x}$$

is étale, so $e(\text{cent}(E, \omega)/b) = 1$. Proposition 1.11 now gives

$$i_E(\omega) = e(\text{cent}(E, \omega)/b) = 1.$$

Taking the contrapositive gives the result. □

Definition 1.13 (Hidden-inertia locus).

$$\mathcal{H}(\mathcal{D}) := \left\{ E \in \text{Ram}^{(1)}(Z/Y) : \begin{array}{l} \exists \omega \in \mathcal{Q}_E, \ i_E(\omega) > 1, \\ \forall \omega \in \mathcal{Q}_E, \ i_E(\omega) > 1 \implies \text{cent}(E, \omega) \in \partial X \end{array} \right\}.$$

Thus $E \in \mathcal{H}(\mathcal{D})$ precisely when nontrivial inertia is visible on a conjugate sheet but every visible center belongs to the deleted boundary.

1.6 Automorphism detection

Theorem 1.14 (Automorphism criterion). *For an arbitrary polynomial self-map $F: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n$, the following are equivalent:*

- (1) F is a polynomial automorphism;
- (2) F is injective on $\mathbb{A}^n(\mathbb{C})$;
- (3) $I_R(F) = I_{\Delta}$;
- (4) $\text{Obs}(F) = 0$.

Proof. Suppose first that F has polynomial inverse $G = (G_1, \dots, G_n)$. Then

$$x_i - y_i = G_i(F(x)) - G_i(F(y)) \in I_R$$

for every i , so $I_{\Delta} \subseteq I_R$. Proposition 1.1 gives equality. Thus (1) implies (3).

Suppose $I_R = I_{\Delta}$, and let $\text{ev}_{(a,b)}: S \rightarrow \mathbb{C}$ be evaluation at (a, b) . If $F(a) = F(b)$, then

$$I_R \subseteq \ker(\text{ev}_{(a,b)}).$$

Therefore $x_i - y_i \in I_{\Delta} = I_R$ gives $a_i - b_i = 0$ for every i . Thus $a = b$, proving (3) $\Rightarrow$ (2). The Ax–Grothendieck theorem [Ax69] gives (2) $\Rightarrow$ (1). Finally, Proposition 1.2 gives (3) $\iff$ (4). $\square$

Corollary 1.15 (The Jacobian conjecture as obstruction vanishing). *For every $n \geq 1$, the n -dimensional Jacobian conjecture is equivalent to each of the statements*

$$\forall F: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n, \quad \det JF \in \mathbb{C}^{\times} \implies \text{Obs}(F) = 0,$$

and

$$\forall F: \mathbb{A}_{\mathbb{C}}^n \rightarrow \mathbb{A}_{\mathbb{C}}^n, \quad \det JF \in \mathbb{C}^{\times} \implies I_R(F) = I_{\Delta}.$$

Proof. Apply Theorem 1.14 to every Keller map. Its equivalent conditions (1), (3), and (4) give the two displayed formulations. $\square$

1.7 Generic degree one

Corollary 1.16 (Generic degree-one descent). *If*

$$L = K \quad \text{and} \quad A \text{ is flat over } B,$$

then

$$\text{Obs}(F) = 0.$$

Proof. When $L = K$, Theorem 1.6 identifies the base-changed diagonal with tensor multiplication:

$$\ker(1 \otimes \bar{\mu}_F) \cong \ker\left(K \otimes_K K \xrightarrow{\mu_K} K\right) = 0.$$

The fiber-product identification of Theorem 1.3 is

$$C_F \xrightarrow{\sim} A \otimes_B A.$$

Since A is flat over B , the tensor product $A \otimes_B A$, and therefore C_F , is flat over B . The localization map

$$\iota_C: C_F \hookrightarrow K \otimes_B C_F$$

is therefore injective, and

$$\iota_C(\ker \bar{\mu}_F) \subseteq \ker(1 \otimes \bar{\mu}_F) = 0.$$

Thus $\ker \bar{\mu}_F = 0$. Proposition 1.2 identifies

$$\ker(\bar{\mu}_F) = \text{Obs}(F) = I_\Delta/I_R,$$

so $\text{Obs}(F) = 0$. □

Lean correspondence. `CollisionIdeals/CollisionDiagonal.lean`, `CollisionIdeals/OffDiagonalScheme.lean`, `CollisionIdeals/Secant.lean`, `CollisionIdeals/GenericCollisionDiagonal.lean`, `CollisionIdeals/KellerCollisionModel.lean`, and `CollisionIdeals/JacobianConjecture.lean`. The Lean form of Theorem 1.6 takes surjectivity of θ as a parameter; the proof above derives it from generic finiteness.

2 Planar vanishing

We now specialize the dimension-independent collision, normalization, and inertia constructions of Section 1 to dimension two.

Let

$$F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \longrightarrow \mathbb{A}_{\mathbb{C}}^2$$

be a polynomial map satisfying $\det JF = c \in \mathbb{C}^\times$.

2.1 The planar secant projector

Let $\Delta = V(I_\Delta) \subseteq \text{Spec } S$. For $a \in S$, write $a|_\Delta$ for its image under $S \rightarrow S/I_\Delta$, and apply this notation entrywise to matrices. Let

$$\mathbf{d} = \begin{pmatrix} x_1 - y_1 \\ x_2 - y_2 \end{pmatrix}, \quad \mathbf{r} = \begin{pmatrix} P(x) - P(y) \\ Q(x) - Q(y) \end{pmatrix}.$$

For $H \in \mathbb{C}[x_1, x_2]$, let

$$D_1 H = \frac{H(x_1, x_2) - H(y_1, x_2)}{x_1 - y_1}, \quad D_2 H = \frac{H(y_1, x_2) - H(y_1, y_2)}{x_2 - y_2}.$$

Both quotients belong to S . Let

$$M_F = \begin{pmatrix} D_1 P & D_2 P \\ D_1 Q & D_2 Q \end{pmatrix}.$$

Then

$$\mathbf{r} = M_F \mathbf{d}, \quad M_F|_\Delta = JF.$$

Let $\delta_F = \det(M_F)$.

Proposition 2.1 (Planar secant annihilation). *The secant determinant satisfies*

$$\delta_F I_\Delta \subseteq I_R, \quad \delta_F|_\Delta = c.$$

Proof. Multiplying $\mathbf{r} = M_F \mathbf{d}$ by $\text{adj}(M_F)$ gives

$$\delta_F \mathbf{d} = \text{adj}(M_F) \mathbf{r}.$$

$$\mathbf{r} \in I_R^{\oplus 2}, \quad \text{adj}(M_F) \in M_2(S) \implies \text{adj}(M_F) \mathbf{r} \in I_R^{\oplus 2}.$$

Therefore

$$\delta_F \mathbf{d} \in I_R^{\oplus 2},$$

or equivalently,

$$\forall j \in \{1, 2\}, \quad \delta_F(x_j - y_j) \in I_R.$$

Hence $\delta_F I_\Delta \subseteq I_R$. On the diagonal,

$$(D_j F_i)|_\Delta = \frac{\partial F_i}{\partial x_j},$$

so $M_F|_\Delta = JF$ and $\delta_F|_\Delta = \det JF = c$. □

Let $\bar{\delta}_F$ be the class of δ_F in C_F , and define

$$q_F := 1 - c^{-1} \bar{\delta}_F.$$

Theorem 2.2 (Planar collision-sheet decomposition). *The collision scheme decomposes as*

$$R_F \cong \Delta_{X/Y}(X) \amalg R_F^\circ, \quad C_F \cong A \times C_F^\circ.$$

The off-diagonal ideal satisfies

$$I_{\text{off}} = I_R + (\delta_F) = I_R : I_\Delta^\infty.$$

Proof. By Proposition 1.2,

$$J := \ker(\bar{\mu}_F) = I_\Delta / I_R = \text{Obs}(F).$$

Proposition 2.1 gives

$$\bar{\delta}_F J = 0, \quad \bar{\delta}_F - c \in J.$$

Theorem 1.5, applied to these two relations and $q_F = 1 - c^{-1} \bar{\delta}_F$, yields

$$q_F^2 = q_F, \quad J = C_F q_F, \quad \text{Ann}_{C_F}(J) = C_F(1 - q_F), \quad J = 0 \iff q_F = 0.$$

Because $1 - q_F = c^{-1} \bar{\delta}_F$,

$$\frac{I_R : I_\Delta}{I_R} = \text{Ann}_{C_F}(J) = C_F(1 - q_F) = C_F \bar{\delta}_F = \frac{I_R + (\delta_F)}{I_R}.$$

Hence

$$I_{\text{off}} = I_R : I_\Delta = I_R + (\delta_F).$$

Idempotence gives $J^m = C_F q_F = J$ for every $m \geq 1$. Therefore

$$\frac{I_R : I_\Delta^m}{I_R} = \text{Ann}_{C_F}(J^m) = \text{Ann}_{C_F}(J) = \frac{I_R : I_\Delta}{I_R}.$$

Thus

$$I_R : I_\Delta^\infty = \bigcup_{m \geq 1} (I_R : I_\Delta^m) = I_R : I_\Delta.$$

The idempotent q_F also gives a ring isomorphism

$$C_F \xrightarrow{\sim} C_F(1 - q_F) \times C_F q_F, \quad a \mapsto (a(1 - q_F), a q_F),$$

whose inverse is addition. The two factors satisfy

$$C_F(1 - q_F) \cong C_F / C_F q_F = C_F / J \xrightarrow[\bar{\mu}_F]{\sim} A$$

and

$$C_F q_F \cong C_F / C_F(1 - q_F) \cong S / (I_R + (\delta_F)) = C_F^\circ.$$

Thus $C_F \cong A \times C_F^\circ$. Since the spectrum of a product is the disjoint union of the two spectra, this isomorphism dualizes to

$$R_F \cong \Delta_{X/Y}(X) \amalg R_F^\circ.$$

□

Corollary 1.4 states that

$$R_F^\circ = \emptyset \iff C_F^\circ = 0 \iff \text{Obs}(F) = 0 \iff I_R = I_\Delta.$$

The preceding theorem identifies

$$C_F^\circ = S / (I_R + (\delta_F)),$$

and Theorem 1.5 gives $\text{Obs}(F) = 0 \iff q_F = 0$. Hence

$$q_F = 0 \iff I_R + (\delta_F) = S \iff R_F^\circ = \emptyset \iff I_R = I_\Delta.$$

In particular, the secant determinant cuts out the off-diagonal factor:

$$R_F^\circ = \text{Spec}(S / (I_R + (\delta_F))).$$

2.2 Boundary exclusion and inertia

Let

$$A = \mathbb{C}[x_1, x_2], \quad B = \mathbb{C}[P, Q] \subseteq A, \quad K = \mathbb{C}(P, Q), \quad L = \mathbb{C}(x_1, x_2),$$

and let $X = \text{Spec } A$ and $Y = \text{Spec } B$.

Proposition 2.3 (Keller local geometry).

$$\mathbb{C}[u, v] \xrightarrow{\sim} B, \quad u \mapsto P, \quad v \mapsto Q.$$

Moreover,

$$X \xrightarrow{f_F} Y \text{ is étale,} \quad A \text{ is flat over } B, \quad [L : K] < \infty \text{ and } L/K \text{ is separable.}$$

Proof. The identity

$$dP \wedge dQ = \left(\frac{\partial P}{\partial x_1} \frac{\partial Q}{\partial x_2} - \frac{\partial P}{\partial x_2} \frac{\partial Q}{\partial x_1} \right) dx_1 \wedge dx_2 = \det(JF) dx_1 \wedge dx_2 = c dx_1 \wedge dx_2 \neq 0$$

shows that P, Q are algebraically independent. Thus $B \cong \mathbb{C}[u, v]$ and $Y \cong \mathbb{A}_{\mathbb{C}}^2$. Moreover,

$$A \cong B[U, V]/(P(U, V) - u, Q(U, V) - v),$$

whose relative Jacobian determinant is $c \in A^\times$. The Jacobian criterion therefore makes $B \rightarrow A$ étale. In particular it is flat and quasi-finite. Since f_F is dominant and quasi-finite, its generic fiber is zero-dimensional, so $[L : K] < \infty$. The morphism remains étale at the generic point; hence the finite extension L/K is separable. $\square$

Fix an algebraic closure $\overline{K} \supseteq L$, and let

$$\begin{aligned} N &:= K(\sigma(L) : \sigma \in \text{Hom}_K(L, \overline{K})), & K &\subseteq L \subseteq N \subseteq \overline{K}, \\ \overline{A} &:= \{a \in L : a \text{ is integral over } B\}, & A_N &:= \{a \in N : a \text{ is integral over } B\}, \\ \overline{X} &:= \text{Spec } \overline{A}, & Z &:= \text{Spec } A_N. \end{aligned}$$

Thus N/K is the normal closure of L/K .

Proposition 2.4 (Planar finite-cover diagram).

$$Z \xrightarrow[\text{finite}]{\pi} \overline{X} \xrightarrow[\text{finite}]{\bar{\nu}} Y, \quad \nu = \bar{\nu} \circ \pi,$$

and

$$X \xrightarrow[\text{open}]{j} \overline{X}, \quad \bar{\nu} \circ j = f_F.$$

Proof. Since $B \cong \mathbb{C}[u, v]$ is a finitely generated normal $\mathbb{C}$ -algebra, its integral closures in the finite extensions L/K and N/K are finite over B . The inclusion $L \subseteq N$ gives $\overline{A} \subseteq A_N$, so

$$B \subseteq \overline{A} \subseteq A_N.$$

The morphism $\bar{\nu}$ is finite because $\overline{A}$ is a finite B -module. If $a_1, \dots, a_r$ generate A_N over B , then they also generate A_N over $\overline{A}$, since $B \subseteq \overline{A}$. Hence π is finite.

For $a \in \overline{A}$, integrality over $B \subseteq A$ and the normality of $A = \mathbb{C}[x_1, x_2]$ give $a \in A$. Thus

$$B \subseteq \overline{A} \subseteq A,$$

and $\text{Frac}(\overline{A}) = L$: clearing denominators in an algebraic equation over $K = \text{Frac}(B)$ shows that for every $\ell \in L$, some nonzero $b \in B$ makes $b\ell$ integral over B . Thus $\ell = (b\ell)/b \in \text{Frac}(\overline{A})$. The inclusion $\overline{A} \subseteq A$ therefore induces a dominant birational morphism $j : X \rightarrow \overline{X}$.

The morphism j is quasi-finite: because $\bar{\nu} \circ j = f_F$, each fiber of j is contained in a fiber of the quasi-finite morphism f_F . It is separated because it is a morphism between affine schemes. Since $\overline{X}$ is normal, Zariski's Main Theorem makes j an open immersion. $\square$

Let

$$\begin{aligned} \mathcal{D}_F &= \left(Z \xrightarrow{\pi} \overline{X} \xrightarrow{\bar{\nu}} Y, j : X \hookrightarrow \overline{X} \right), & \partial X &= \overline{X} \setminus j(X), \\ G &= \text{Gal}(N/K), & H &= \text{Gal}(N/L). \end{aligned}$$

For $E \in \text{Ram}^{(1)}(Z/Y)$ and $\omega \in \mathcal{Q}_E$, Proposition 1.11 gives

$$e(\text{cent}(E, \omega)/\nu(E)) = i_E(\omega).$$

Since $X \rightarrow Y$ is étale by Proposition 2.3, Proposition 1.12 gives

$$i_E(\omega) > 1 \implies \text{cent}(E, \omega) \in \partial X.$$

Thus Definition 1.13 specializes to

$$\mathcal{H}(\mathcal{D}_F) = \left\{ E \in \text{Ram}^{(1)}(Z/Y) : \begin{array}{l} \exists \omega \in \mathcal{Q}_E, \ i_E(\omega) > 1, \\ \forall \omega \in \mathcal{Q}_E, \ i_E(\omega) > 1 \implies \text{cent}(E, \omega) \in \partial X \end{array} \right\}.$$

2.3 Hidden inertia and divisorial ramification

Corollary 2.5 (Planar divisorial ramification criterion).

$$\mathcal{H}(\mathcal{D}_F) = \emptyset \iff \text{Ram}^{(1)}(Z/Y) = \emptyset.$$

Proof. The reverse implication follows from $\mathcal{H}(\mathcal{D}_F) \subseteq \text{Ram}^{(1)}(Z/Y)$. Conversely, suppose that $\mathcal{H}(\mathcal{D}_F) = \emptyset$ and let $E \in \text{Ram}^{(1)}(Z/Y)$. Then $I_E \neq 1$. Since N is the normal closure of L/K , $\text{core}_G(H) = 1$, and Proposition 1.10 gives

$$\exists \omega \in \mathcal{Q}_E, \quad i_E(\omega) > 1.$$

For every $\omega' \in \mathcal{Q}_E$, Proposition 1.12 gives

$$i_E(\omega') > 1 \implies \text{cent}(E, \omega') \in \partial X.$$

Hence $E \in \mathcal{H}(\mathcal{D}_F)$, a contradiction. □

2.4 Collapse of the normal closure

Theorem 2.6 (Purity of the branch locus, specialized). *Let $Z \rightarrow \mathbb{A}_{\mathbb{C}}^2$ be a finite dominant generically separable morphism with Z integral and normal. If $\text{Ram}^{(1)}(Z/\mathbb{A}_{\mathbb{C}}^2) = \emptyset$, then the morphism is étale.*

Proof. At every height-one point of Z , the absence of divisorial ramification gives ramification index one. The corresponding residue field extension is finite and, in characteristic zero, separable. Hence the morphism is unramified at every height-one point. Purity of the branch locus now implies that it is étale everywhere [The26]. □

Theorem 2.7 (Finite-étale rigidity of the affine plane). *Every connected finite étale cover of $\mathbb{A}_{\mathbb{C}}^2$ is trivial [GR71, Exposé XII].*

Corollary 2.8 (Triviality of the planar normal closure). *If*

$$\text{Ram}^{(1)}(Z/Y) = \emptyset,$$

then

$$N = K \quad \text{and hence} \quad L = K.$$

Proof. By Proposition 2.3, $Y \cong \mathbb{A}_{\mathbb{C}}^2$. Since N/K is finite Galois, $\nu: Z \rightarrow Y$ is generically separable; hence Theorem 2.6 makes it finite étale. The inclusion

$$A_N \subset N \text{ is a domain,} \quad \text{so} \quad Z = \text{Spec } A_N \text{ is connected.}$$

Theorem 2.7 therefore makes $\nu: Z \rightarrow Y$ an isomorphism. Taking function fields gives $N = K$. Finally, $K \subseteq L \subseteq N$ implies $L = K$. $\square$

2.5 Vanishing of the off-diagonal factor

Theorem 2.9 (Planar Vanishing). *For every polynomial map $F = (P, Q): \mathbb{A}_{\mathbb{C}}^2 \rightarrow \mathbb{A}_{\mathbb{C}}^2$ satisfying $\det JF = c \in \mathbb{C}^\times$,*

$$\mathcal{H}(\mathcal{D}_F) = \emptyset \implies \begin{array}{ll} I_R = I_\Delta, & \text{Obs}(F) = 0, \\ q_F = 0, & R_F^\circ = \emptyset. \end{array}$$

Proof. Corollary 2.5 gives $\text{Ram}^{(1)}(Z/Y) = \emptyset$, and Corollary 2.8 gives $L = K$. Proposition 2.3 gives that A is flat over B . Corollary 1.16 states

$$L = K, \quad A \text{ flat over } B \implies \text{Obs}(F) = 0,$$

so $\text{Obs}(F) = 0$.

Corollary 1.4 and Theorem 1.5 state

$$\begin{array}{ccccccc} R_F^\circ = \emptyset & \iff & C_F^\circ = 0 & \iff & \text{Obs}(F) = 0 & \iff & I_R = I_\Delta, \\ & & & & \text{Obs}(F) = 0 & \iff & q_F = 0. \end{array}$$

All four conclusions follow. $\square$

Corollary 2.10 (Conditional Jacobian conjecture in dimension two). *If $\mathcal{H}(\mathcal{D}_F) = \emptyset$ for every planar Keller map F , then every planar Keller map is a polynomial automorphism. Equivalently, the two-dimensional Jacobian conjecture holds.*

Proof. The automorphism criterion, Theorem 1.14, states that every polynomial self-map of $\mathbb{A}_{\mathbb{C}}^2$ satisfies

$$I_R = I_\Delta \iff F \text{ is a polynomial automorphism.}$$

Planar Vanishing supplies the left-hand side for every planar Keller map under the stated condition. $\square$

Lean correspondence. The conditional planar chain is collected in `CollisionIdeals/Planar.lean`. The development takes the following established results as explicit inputs: planar Ax–Grothendieck, branch purity, and finite-étale rigidity of $\mathbb{A}_{\mathbb{C}}^2$. It also takes a supplied `PlanarKellerCollisionModelF` and the no-hidden-inertia hypothesis; constructing the model and proving that hypothesis remain separate formalization tasks.

3 The cubic S_3 obstruction in dimension three

We now specialize the dimension-independent constructions of Section 1 to dimension three and isolate the separable nonnormal cubic case.

Let $S_3 = \text{Perm}(\{1, 2, 3\})$ be the symmetric group on three letters.

Proposition 3.1 (Dimension-three counterexample criterion). *For a polynomial map*

$$F: \mathbb{A}_{\mathbb{C}}^3 \longrightarrow \mathbb{A}_{\mathbb{C}}^3,$$

consider the following statements:

- (1) $\det JF \in \mathbb{C}^\times$;
- (2) F is not a polynomial automorphism;
- (3) $R_F^\circ \neq \emptyset$.

Statements (2) and (3) are equivalent. Consequently, F is a dimension-three Jacobian-conjecture counterexample exactly when (1) and (3) hold.

Proof. By Corollary 1.4 and Theorem 1.14,

$$R_F^\circ = \emptyset \iff I_R = I_\Delta \iff F \in \text{Aut}_{\text{poly}}(\mathbb{A}_{\mathbb{C}}^3).$$

Negating this equivalence proves (2) $\iff$ (3). Combining this with (1) gives the final assertion. $\square$

3.1 The cubic extension and its normal closure

Let

$$F = (F_1, F_2, F_3): \mathbb{A}_{\mathbb{C}}^3 \longrightarrow \mathbb{A}_{\mathbb{C}}^3$$

be dominant and generically finite. Let

$$A = \mathbb{C}[x_1, x_2, x_3], \quad B = \mathbb{C}[F_1, F_2, F_3] \subseteq A, \quad K = \text{Frac}(B), \quad L = \text{Frac}(A),$$

and let

$$\theta: K \otimes_B A \longrightarrow L, \quad \lambda \otimes a \longmapsto \lambda a,$$

be the isomorphism of Theorem 1.6. Suppose that L/K is separable and nonnormal, and choose a power basis with generator α and degree three:

$$L = K(\alpha), \quad [L : K] = 3.$$

If $m_\alpha \in K[T]$ is the minimal polynomial of α , let $h_\alpha \in L[T]$ be determined by

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \quad \text{in } L[T].$$

Let N/K be the normal closure of L/K , containing L through the marked inclusion. Let

$$G = \text{Gal}(N/K), \quad H = \text{Gal}(N/L).$$

Recall from Section 1.5 that

$$G/H \xrightarrow{\sim} \text{Hom}_K(L, N), \quad gH \mapsto g|_L, \quad \text{core}_G(H) = 1.$$

The final equality follows because N is the normal closure of L/K ; equivalently, the action of G on the conjugate embeddings of L is faithful.

For $Y = \text{Spec } B$, the finite-normalization construction of Section 1.5 gives

$$Z = \text{Norm}_N(Y) \longrightarrow \overline{X} = \text{Norm}_L(Y) \longrightarrow Y.$$

If $E \in Z^{(1)}$, with decomposition group D_E and inertia group I_E , then its conjugate centers are indexed by $\mathcal{Q}_E = D_E \backslash G/H$. For $\omega = D_E gH$, Proposition 1.11 states

$$e(\text{cent}(E, \omega)/\nu(E)) = i_E(\omega) = [I_E : I_E \cap gHg^{-1}].$$

The following proposition is the standard residual-factor description of a separable nonnormal cubic extension. We include its short proof to fix the marked identification with N .

Proposition 3.2 (Cubic residual-factor identification). *The polynomial h_α is irreducible over L , and there is an L -algebra isomorphism*

$$\varphi: L[T]/(h_\alpha) \xrightarrow{\sim}_L N.$$

Consequently,

$$\Psi: L \otimes_K L \xrightarrow{\sim}_L L \times N, \quad \text{pr}_1 \circ \Psi = \mu_L.$$

Proof. Write

$$m_\alpha(T) = T^3 + a_2T^2 + a_1T + a_0.$$

If the quadratic h_α were reducible over L , it would have a root $\beta \in L$. Separability gives $\beta \neq \alpha$, and the third root would be

$$\gamma = -a_2 - \alpha - \beta \in L.$$

Thus m_α would split over L . Since $L = K(\alpha)$, the extension L/K would then be the splitting field of the separable polynomial m_α , hence normal. This contradicts the assumed nonnormality, so h_α is irreducible.

Let β denote the image of T in $L[T]/(h_\alpha)$. This field contains the three roots

$$\alpha, \quad \beta, \quad -a_2 - \alpha - \beta$$

of m_α , and is generated over L by β . It is therefore the splitting field of m_α , which is the normal closure N of L/K . Choose the resulting L -algebra isomorphism

$$\varphi: L[T]/(h_\alpha) \xrightarrow{\sim}_L N.$$

Let

$$\Phi_\alpha: L \otimes_K L \xrightarrow{\sim}_L L \times L[T]/(h_\alpha)$$

be the marked-root decomposition of Theorem 1.7. It satisfies

$$\text{pr}_1 \circ \Phi_\alpha = \mu_L.$$

Composing its second factor with φ , let

$$\Psi = (\text{id}_L \times \varphi) \circ \Phi_\alpha.$$

Since $\text{id}_L \times \varphi$ leaves the first factor unchanged, $\text{pr}_1 \circ \Psi = \mu_L$. □

Theorem 3.3 (Galois group of a nonnormal cubic extension).

$$\mathrm{Gal}(N/K) \cong S_3.$$

Proof. Proposition 3.2 identifies N with the degree-two extension $L[T]/(h_\alpha)$ of L . Hence

$$|H| = [N : L] = 2.$$

The normal-closure identities recalled above give

$$\mathrm{core}_G(H) = 1, \quad [G : H] = [L : K] = 3.$$

Hence the action of G on the three cosets of H gives

$$\rho: G \longrightarrow \mathrm{Perm}(G/H) \cong S_3, \quad \ker \rho = \mathrm{core}_G(H) = 1.$$

Core-freeness therefore makes ρ injective, so

$$|G| \leq |S_3| = 6.$$

The index formula gives

$$|G| = [G : H]|H| = 3 \cdot 2 = 6.$$

Thus $|G| = 6$, and the injective homomorphism ρ is an isomorphism $G \cong S_3$. $\square$

3.2 The cubic off-diagonal factor

Retain the cubic data of Section 3.1, and suppose in addition that F is Keller. Thus

$$B = \mathbb{C}[F_1, F_2, F_3] \subseteq A = \mathbb{C}[x_1, x_2, x_3], \quad K = \mathrm{Frac}(B) \subseteq L = \mathrm{Frac}(A) = K(\alpha) \subseteq N,$$

where L/K is separable, nonnormal, and of degree three, and N/K is its normal closure. Let

$$\theta: K \otimes_B A \xrightarrow{\sim} L, \quad \lambda \otimes a \longmapsto \lambda a.$$

Corollary 3.4 (Generic cubic S_3 collision). *There is a K -algebra isomorphism*

$$\Psi_F: K \otimes_B C_F \xrightarrow{\sim}_K L \times N, \quad \mathrm{pr}_1 \circ \Psi_F = \theta \circ (1 \otimes \bar{\mu}_F).$$

It satisfies

$$\Psi_F(\ker(\theta \circ (1 \otimes \bar{\mu}_F))) = 0 \times N, \quad \mathrm{Gal}(N/K) \cong S_3.$$

Consequently,

$$\mathrm{Obs}(F) \neq 0, \quad I_R \subsetneq I_\Delta, \quad R_F^\circ \neq \emptyset, \quad F \notin \mathrm{Aut}_{\mathrm{poly}}(\mathbb{A}_{\mathbb{C}}^3).$$

Proof. Theorem 1.6 gives a K -algebra isomorphism

$$\Phi_F: K \otimes_B C_F \xrightarrow{\sim}_K L \otimes_K L$$

such that

$$\mu_L \circ \Phi_F = \theta \circ (1 \otimes \bar{\mu}_F).$$

Proposition 3.2 gives

$$\Psi: L \otimes_K L \xrightarrow{\sim} L \times N, \quad \text{pr}_1 \circ \Psi = \mu_L.$$

After restricting Ψ from L to K , let

$$\Psi_F = \Psi \circ \Phi_F.$$

It follows that

$$\text{pr}_1 \circ \Psi_F = \theta \circ (1 \otimes \bar{\mu}_F), \quad \Psi_F(\ker(\theta \circ (1 \otimes \bar{\mu}_F))) = 0 \times N \neq 0.$$

Suppose that the diagonal-evaluation homomorphism $\bar{\mu}_F: C_F \rightarrow A$ were injective. Since K is a localization of B , it is flat over B , so $1 \otimes \bar{\mu}_F$ would also be injective. Theorem 1.6 shows that θ is an isomorphism. Therefore $\theta \circ (1 \otimes \bar{\mu}_F)$ would be injective, contradicting the nonzero kernel $0 \times N$ above. Hence $\ker(\bar{\mu}_F) \neq 0$, and Proposition 1.2 gives $\text{Obs}(F) \neq 0$. Proposition 1.1 shows that $I_R \subseteq I_\Delta$ strict, and Corollary 1.4 gives $R_F^\circ \neq \emptyset$.

Finally, Theorem 3.3 identifies $\text{Gal}(N/K)$ with S_3 . If F were a polynomial automorphism, Theorem 1.14 would give $\text{Obs}(F) = 0$, contrary to $\text{Obs}(F) \neq 0$. Since F is Keller, it is a dimension-three Jacobian-conjecture counterexample. $\square$

For the same cubic map and field tower, let

$$X = \text{Spec } A, \quad Y = \text{Spec } B, \quad G = \text{Gal}(N/K), \quad H = \text{Gal}(N/L).$$

Suppose there is a diagram over Y

$$\mathcal{D} = \left(Z = \text{Norm}_N(Y) \xrightarrow{\pi} \bar{X} = \text{Norm}_L(Y) \xrightarrow{\bar{\nu}} Y, j: X \hookrightarrow \bar{X} \right)$$

in which π and $\bar{\nu}$ are finite, j is an open immersion, $\bar{\nu} \circ j = f_F$, and $f_F: X \rightarrow Y$ is étale. Let

$$\partial X = \bar{X} \setminus j(X), \quad G \cong S_3$$

by Theorem 3.3.

Corollary 3.5 (Hidden inertia in the S_3 -normalization).

$$\text{Ram}^{(1)}(Z/Y) \subseteq \mathcal{H}(\mathcal{D}).$$

Proof. Fix

$$E \in \text{Ram}^{(1)}(Z/Y).$$

By the characteristic-zero description of divisorial ramification, $I_E \neq 1$. Because N is the normal closure of L/K , $\text{core}_G(H) = 1$. Proposition 1.10 states in this case that

$$\exists \omega \in \mathcal{Q}_E, \quad i_E(\omega) > 1.$$

Moreover, Proposition 1.12, applied to the étale map f_F , states that for every $\omega' \in \mathcal{Q}_E$,

$$i_E(\omega') > 1 \implies \text{cent}(E, \omega') \in \partial X.$$

These are exactly the existence and boundary conditions in Definition 1.13; hence $E \in \mathcal{H}(\mathcal{D})$. Since E was arbitrary, the stated inclusion follows. $\square$

Remark (Ordered roots). Let

$$m_\alpha(T) = T^3 + a_2T^2 + a_1T + a_0$$

have roots $\alpha_1, \alpha_2, \alpha_3$ in N . For distinct i, j, k ,

$$\alpha_k = -a_2 - \alpha_i - \alpha_j, \quad K(\alpha_i, \alpha_j) = K(\alpha_1, \alpha_2, \alpha_3) = N.$$

Thus the off-diagonal factor, which marks two distinct roots, is already the ordered-root normal closure.

Lean correspondence. `CollisionIdeals/ComplexThree/CubicGaloisGroup.lean`, `CollisionIdeals/ComplexThree/S3Collision.lean`, `CollisionIdeals/ComplexThree/OffDiagonal.lean`, `CollisionIdeals/ComplexThree/CubicS3.lean`, and `CollisionIdeals/ComplexThree/JacobianConjecture.lean`. The Lean cubic witness takes the residual-field isomorphism and $H \neq 1$ as explicit inputs; Proposition 3.2 derives them here from nonnormality. Hidden inertia is formalized pointwise at a supplied ramified divisor.

4 Conclusion

The obstruction ideal $\text{Obs}(F)$ vanishes exactly when the scheme-theoretic collision relation $X \times_Y X$ equals the diagonal. In dimension two, $\mathcal{H}(\mathcal{D}_F) = \emptyset$ implies $\text{Obs}(F) = 0$; in the cubic dimension-three class, its nonzero generic factor is the ordered-root S_3 -normal closure. Both statements arise from the same collision, off-diagonal, and conjugate-sheet constructions.

Under the respective conditions of Sections 2 and 3, and with $\text{Ram}^{(1)}(Z/Y) \neq \emptyset$ in the cubic normalization, the contrast is

dimension two	dimension three
hidden inertia excluded	hidden inertia in the S_3 -cover realized
$\text{Obs}(F) = 0$	$\text{Obs}(F) \neq 0$
$I_R = I_\Delta$	$I_R \subsetneq I_\Delta$
F is a polynomial automorphism	F is not a polynomial automorphism.

Motivating example. The explicit three-dimensional polynomial counterexample announced by Levent Alpöge [Alp26] and independently formalized in Isabelle/HOL by Ramos, Hulak, and de Queiroz [RHdQ26] motivated the cubic S_3 comparison. The formal verification establishes a constant nonzero Jacobian determinant and distinct rational points with the same image. That particular map is not used in the arguments above: the dimension-three result is the structural implication from the stated cubic and marked-normal-closure conditions.

The formalization uses Lean 4 and Mathlib [dMU21, The20].

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